Rossmoyne Senior High School

Semester One Examination, 2019

Question/Answer booklet

MATHEMATICS APPLICATIONS UNIT 1

Section Two: Calculator-assumed

SOLUTIONS

Student number: In figures

In words ______________________________

Your name ______________________________

Time allowed for this section

Reading time before commencing work: ten minutes
Working time: one hundred minutes

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet
Formula sheet (retained from Section One)

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators approved for use in this examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	8	8	50	52	35
Section Two: Calculator-assumed	13	13	100	98	65
				Total	100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.

2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.

3. You must be careful to confine your answer to the specific question asked and to follow any instructions that are specified to a particular question.

4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.

5. It is recommended that you do not use pencil, except in diagrams.

6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.

7. The Formula sheet is not to be handed in with your Question/Answer booklet.

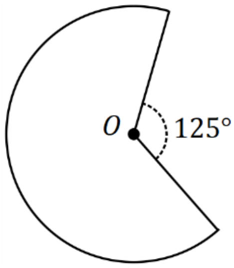
Section Two: Calculator-assumed **65% (98 Marks)**

This section has **thirteen (13)** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 9 **(5 marks)**

A sector of a circle of radius 30 cm is shown below.



(a) Show that the perimeter of the sector is 183 cm, when rounded to the nearest cm. (3 marks)

Solution
Interior angle: $360 - 125 = 235°$ Arc length: $L = 2\pi(30) \times \frac{235}{360} = 123$ Perimeter: $P = 123 + 30 + 30 = 183$ cm
Specific behaviours
✓ interior angle ✓ arc length ✓ total perimeter

(b) Determine the area of the sector. (2 marks)

Solution
Area: $A = \pi(30)^2 \times \frac{235}{360}$ $A = 2827 \times \frac{235}{360} \approx 1\,846$ cm^2
Specific behaviours
✓ area of circle ✓ area of sector

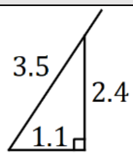
Question 10 **(8 marks)**

(a) A triangle has sides of length 55 cm, 48 cm and 71 cm. State, with reasons, whether the triangle is right-angled. (2 marks)

Solution
$48^2 + 55^2 = 5329 \neq 5041 = 71^2$ Hence triangle is not right-angled as the side lengths do not satisfy **Pythagoras' theorem**.
Specific behaviours
✓ shows use of $a^2 + b^2 = c^2$ ✓ explanation (must use bolded words)

(b) A 3.5 m long ladder leans against a 2.4 m tall wall with the foot of the ladder 1.1 m away from the base of the wall on level ground.

(i) Draw a sketch of this situation, showing the ladder extending beyond the top of the wall. (2 marks)

Solution

Specific behaviours
✓ sketch with right-angle ✓ includes lengths

(ii) Calculate the distance the ladder extends beyond the top of the wall. (2 marks)

Solution
$\sqrt{1.1^2 + 2.4^2} = 2.64$ $3.5 - 2.64 = 0.86$ m
Specific behaviours
✓ length touching wall ✓ distance extended

(iii) The foot of the ladder is moved along the ground and away from the base of the wall until the top of the ladder is just touching the top of the wall. Determine how far the foot of the ladder was moved. (2 marks)

Solution
Distance from bottom of wall: $\sqrt{3.5^2 - 2.4^2} = 2.55$ Moved $2.55 - 1.1 = 1.45$ m
Specific behaviours
✓ distance from wall ✓ distance moved

Question 11 **(7 marks)**

(a) Calculate the value of q given that $q = \frac{1}{e}$ and $e = \frac{1}{e_1} + \frac{1}{e_2}$ when $e_1 = 3.96$ and $e_2 = 0.44$.

(2 marks)

Solution
$e = \frac{1}{3.96} + \frac{1}{0.44} = \frac{250}{99} = 2.\overline{52}$ $q = 1 \div \frac{250}{99} = 0.396$
Specific behaviours
✓ correct value of e ✓ correct answer

(b) The future value, V, of an annuity can be calculated using the formula below.

$$V = \frac{C}{R}\left(\left(1 + \frac{R}{N}\right)^{NT} - 1\right)$$

Consider an annuity with values $C = \$3\,750,\ R = 0.032,\ N = 12$ and $T = 7$.

(i) Calculate the future value, rounding your answer to the nearest dollar. (3 marks)

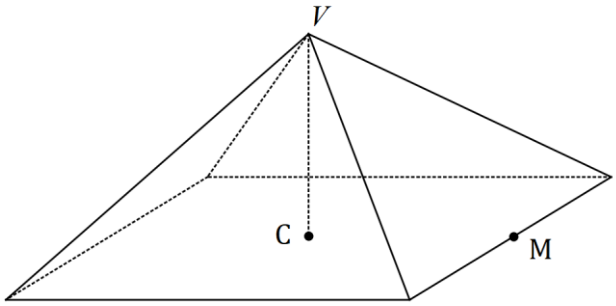
Solution
$V = \frac{3750}{0.032}\left(\left(1 + \frac{0.032}{12}\right)^{12\times 7} - 1\right)$ $V = 29378.68$ $V \approx \$29\,379$ (nearest \$)
Specific behaviours
✓ evidence of correct substitution   ✓ calculates V ✓ value to nearest dollar

(ii) Calculate the change in V when the value of N is changed from 12 to 6. (2 marks)

Solution
$V \approx \$29\,335$ (nearest \$)   Decrease of $29\,379 - 29\,335 = \$44$
Specific behaviours
✓ calculates V ✓ correct decrease

Question 12 **(6 marks)**

A bakery sells a specialty cake in the shape of a square-based pyramid. The sides of the square are all 38 cm and the vertex of the pyramid, V, lies 24 cm directly above the centre of the square base, C. The mid-point of one of the sides of the square is M.



(a) Determine the volume of the cake. (2 marks)

Solution
$V = \frac{1}{3}(38 \times 38) \times 24$ $= 11\ 552\ \text{cm}^2$
Specific behaviours
✓ indicates area of base ✓ correct volume

(b) Calculate the length VM. (2 marks)

Solution
$MC = 38 \div 2 = 19$ $VM = \sqrt{24^2 + 19^2}$ $= 30.6\ \text{cm}$
Specific behaviours
✓ indicates lengths MC and VC ✓ correct distance

(c) The four triangular faces are iced at a cost of 1.7 cents per square cm. Determine the cost of icing the cake, in dollars and cents. (2 marks)

Solution
$A = 4 \times \frac{1}{2} \times 32 \times 30.6$ $= 2326$ $C = 2326 \times 0.017$ $= \$39.55$
Specific behaviours
✓ correct TSA ✓ correct cost

Question 13 **(6 marks)**

A researcher at a weather station uses the following formula to calculate A, the absolute humidity of the air, where R is the relative humidity and T is the air temperature.

$$A = \frac{13.1 \times R \times (1.06)^T}{T + 275}$$

(a) Calculate A when $R = 55$ and $T = 20$. (2 marks)

Solution
$A = \frac{13.1 \times 55 \times (1.06)^{20}}{20 + 275}$ $= 7.83$
Specific behaviours
✓ substitutes correctly   ✓ correct value

The formula for A was used to create the spreadsheet below for various values of R and T.

A		R				
		30	40	50	60	70
T	-5	1.09	P	1.81	2.18	2.54
	0	1.43	1.91	2.38	2.86	Q
	5	1.88	2.50	3.13	3.76	4.38
	10	3.35	3.29	4.12	4.94	5.76

(b) Calculate the value of the entries P and Q shown in the spreadsheet. (2 marks)

Solution
$P = 1.45$ $Q = 3.33$
Specific behaviours
✓ value of P ✓ value of Q

(c) One of the entries in the spreadsheet is incorrect. State the value of T and the value of R for the cell in which the incorrect value lies and calculate the correct value of A. (2 marks)

Solution
$T = 10, \quad R = 30$ $A = 2.47$
Specific behaviours
✓ correct values of T and H ✓ correct value of A

Question 14 **(9 marks)**

The weekly sales of Waywoo phones (the Nine, Plus and Deluxe models) at retail outlets A, B and C is shown in matrix M.

$$M = \begin{array}{c} \\ \\ \\ \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{bmatrix} 70 & 35 & 41 \\ 45 & 25 & 32 \\ 44 & 36 & 31 \end{bmatrix} \end{array} \begin{array}{c} \\ Nine \\ Plus \\ Deluxe \end{array}$$

(a) In total, how many phones were sold by store C? (1 mark)

Solution
$41 + 32 + 31 = 104$
Specific behaviours
✓ correct number

(b) What does the element m_{13} in matrix M represent? (1 mark)

Solution
The number of Nine models sold by store C
Specific behaviours
✓ mentions store and model

(c) Determine the matrix F, where $F = M \times \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, and explain what F shows. (2 marks)

Solution
$F = \begin{bmatrix} 146 \\ 102 \\ 111 \end{bmatrix}$ The total number of each model sold during the week.
Specific behaviours
✓ correct F ✓ correct explanation

(d) Determine the matrix G, where $G = [1 \quad 1 \quad 1] \times M$, and explain what G shows. (2 marks)

Solution
$G = [159 \quad 96 \quad 104]$ The total number of phones sold at each of the stores during the week.
Specific behaviours
✓ correct G ✓ correct explanation

(e) The Nine, Plus and Deluxe models are sold for $\$749$, $\$899$ and $\$979$ respectively.

(i) Use this information to write matrix H that can be multiplied by matrix F to yield a sensible result. (1 mark)

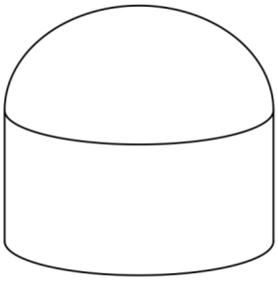
Solution
$H = [749 \quad 899 \quad 979]$
Specific behaviours
✓ correct matrix

(ii) Multiply matrices H and F together and explain what the result shows. (2 marks)

Solution
$H \times F = [309721]$ The total sales income from selling all the phones at all the stores.
Specific behaviours
✓ correct product, as matrix ✓ correct explanation

Question 15 **(6 marks)**

A wood turner crafted solid shape S as shown below - a hemisphere of radius 15 cm atop a cylinder with the same radius and height 18 cm.



(a) The wood turner started out with a block of wood in the shape of a rectangular prism with dimensions $40 \times 35 \times 35$ cm. Determine the volume of wood that was removed from this block to end up with shape S. (3 marks)

Solution
$V_{CYL} = \pi(15)^2(18)$ $= 12\,723$ $V_{HS} = \frac{1}{2} \times \frac{4}{3} \times \pi(15)^3$ $= 7\,069$ $V_{PRISM} = 40 \times 35 \times 35 = 49\,000$ $V_{REMOVED} = 49000 - 12723 - 7069$ $= 29\,208 \text{ cm}^3$
Specific behaviours
✓ cylinder volume ✓ hemisphere volume ✓ correct volume removed

(b) Shape S was then cut vertically into two congruent halves. Determine the area of one of the cut faces. (3 marks)

Solution
$A_{SEMICIRCLE} = \frac{1}{2} \times \pi \times 15^2$ $= 353$ $A_{RECTANGLE} = 30 \times 18$ $= 540$ $A_{CUT} = 353 + 540$ $= 893 \text{ cm}^2$
Specific behaviours
✓ uses correct cross-sections ✓ semi-circle area ✓ correct total

Question 16 **(8 marks)**

(a) An investor pays $\$12\,750$ into a new account offering interest of 3.24% pa, with interest payable every 60 days. The investor closes the account and withdraws the principal and interest just after the first interest payment.

Calculate the amount they withdraw. (3 marks)

Solution
$I = 12750 \times 0.0324 \times 60 \div 365$ $= 67.91$ $12750 + 67.91 = \$12\,817.91$
Specific behaviours
✓ correctly substitutes ✓ calculates interest ✓ correct withdrawal

(b) Another person invested $\$2\,500$ with a bank that offered an interest rate of 6.2% pa.

(i) Determine the value of the investment after two years when interest is compounded annually. (2 marks)

Solution
$A = 2500(1 + 0.062)^2$ $= \$2\,819.61$
Specific behaviours
✓ correctly substitutes ✓ correct value

(ii) The person was advised that they would be better off after two years if they invested the same sum with another bank that was offering an interest rate of 6.1% pa compounded monthly. State, with justification, whether this was good advice. (3 marks)

Solution
$A = 2500\left(1 + \frac{0.061}{12}\right)^{2\times 12}$ $= \$2\,823.51$ Advice was good - they were $\$3.90$ better off.
Specific behaviours
✓ correctly substitutes ✓ correct value ✓ correct conclusion

Question 17 **(12 marks)**

A spreadsheet to track a monthly car budget is shown below, with all amounts in dollars. The figures for months 1, 2 and 3 show the actual expenditure for each item.

Item	Monthly Budget	Month 1	Month 2	Month 3	3-Month Total
Loan payment	344	344	344	344	R
Insurance	215	215	215	255	685
Maintenance	129	0	88	295	383
Fuel	172	195	Q	173	536
Total	P	754	815	1067	S

(a) Determine the values of P, Q, R and S in the spreadsheet. (3 marks)

Solution
$P = 860$ $Q = 536 - (195 + 173) = 168$ $R = 3 \times 344 = 1\,032$ $S = 2\,636$
Specific behaviours
✓ one correct value ✓ at least 3 correct values ✓ all correct values

(b) A motoring organisation recommends allocating no more than 20% of a person's monthly income to running a car. Would a person on a salary of $\$52\,200$ pa and using the total monthly budget figure above meet this target? Justify your answer. (3 marks)

Solution
$52200 \div 12 = 4350$ $\text{Target: } 4350 \times 0.20 = \$870$ Since the budget total is $860 and the recommended maximum is $870, they just meet the target.
Specific behaviours
✓ calculates monthly salary ✓ calculates correct percentage ✓ explains why they meet the target

(c) Over the three months shown, the total actual expenditure for two items exceeded their budgeted figures.

(i) Which items were these? (1 mark)

Solution
Insurance and fuel $(172 \times 3 = 516)$
Specific behaviours
✓ correct items

(ii) Determine which of these two items exceeded their budgeted figure by the largest percentage. (3 marks)

Solution
Insurance: $685 - 3(215) = 685 - 645 = 40$ $$\frac{40}{645} \times 100 = 6.2\%$$ Fuel: $536 - 3(172) = 536 - 516 = 20$ $$\frac{20}{516} \times 100 = 3.9\%$$ Insurance exceeded by the largest percentage.
Specific behaviours
✓ % figure for insurance ✓ % figure for fuel ✓ states correct item

(d) The monthly budget figure for insurance includes a 10% tax. Determine the cost of the insurance without this tax. (2 marks)

Solution
$215 \div 1.10 = \$195.45$
Specific behaviours
✓ indicates appropriate method ✓ correct cost, to nearest cent

Question 18 **(7 marks)**

A shop sold three sizes of the same brand of tomato sauce. The 250 mL bottle cost $\$1.55$, the 475 mL bottle cost $\$2.66$ and the 920 mL bottle cost $\$5.52$.

Retail laws require the shop to display the price per 100 mL for each bottle, to the nearest cent.

(a) Calculate the price per 100 mL for all three sizes of tomato sauce. (3 marks)

Solution
$250: 155 \div 250 \times 100 = 62 \text{ c}$ $475: \ 266 \div 475 \times 100 = 56 \text{ c}$ $920: \ 552 \div 920 \times 100 = 60 \text{ c}$
Specific behaviours
✓ uses unit cost for one size ✓ correct 100 mL price for one size ✓ all three prices correct

(b) Use the unit prices to list the bottle sizes in order of value, from best to worst. (1 mark)

Solution
475 mL, 920 mL, 250 mL
Specific behaviours
✓ correct order

(c) Determine the maximum price, to the nearest cent, for a 725 mL bottle of the same sauce if the shop wanted it to be better value than the other three sizes in terms of unit price. (2 marks)

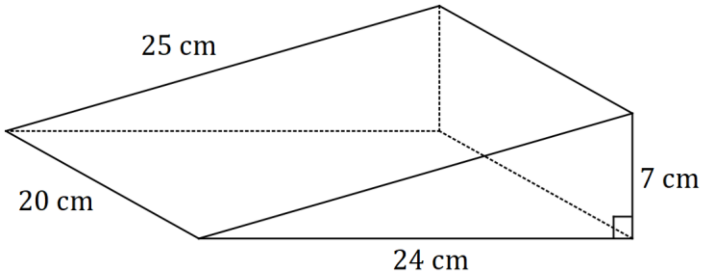
Solution
$725 \div 100 \times 56 = 406 \text{ c}$ To be best value, price must be less, so $\$4.05$.
Specific behaviours
✓ calculates price to be equal best ✓ correct price

(d) Suggest one reason that a shopper may prefer a size other than that which offers the best value in terms of unit price. (1 mark)

Solution
- avoid wastage - don't have enough money, etc, etc
Specific behaviours
✓ sensible reason

Question 19 **(8 marks)**

A right-triangular prism has dimensions shown in the diagram below.



(a) Determine the total surface area of the prism. (3 marks)

Solution
Triangular ends: $2 \times \frac{1}{2} \times 24 \times 7 = 168$ Rectangles: $20 \times (7 + 24 + 25) = 1120$ TSA: $168 + 1120 = 1\,288$ cm^2
Specific behaviours
✓ indicates area of triangle ✓ indicates area of rectangle ✓ correct TSA

(b) A quarter-size model is made of the prism, i.e. using a scale of $1:4$. Calculate the total surface area of this model. (2 marks)

Solution
Area scale factor: $\left(\frac{1}{4}\right)^2 = \frac{1}{16}$ Area: $1288 \times \frac{1}{16} = 80.5$ cm^2
Specific behaviours
✓ indicates area scale factor ✓ correct area

(c) Determine the volume of the prism. (1 mark)

Solution
Volume: $\frac{1}{2} \times 24 \times 7 \times 20 = 1\,680$ cm^3
Specific behaviours
✓ correct volume

(d) Another scale model is made of the prism, so that the dimensions of the sloping face are now 4 cm by 5 cm. Calculate the volume of this scale model. (2 marks)

Solution
Scale factor: $\frac{4}{20} = \frac{1}{5}$. Volume sf $\left(\frac{1}{5}\right)^3 = \frac{1}{125}$ Volume: $1680 \times \frac{1}{125} = 13.44$ cm^3
Specific behaviours
✓ indicates scale factor for length and volume ✓ correct volume

Question 20 **(8 marks)**

In the country of Wealthyland, the government will pay allowances to families with children. The following tables represent the amount of allowance a family can receive.

Families with 1 child meeting the criteria	
Combined annual income	**Allowance for the year**
Up to \$54 000	\$5100
\$54 000 to \$73 000	\$5100 less 10 cents for each \$1 that annual income exceeds \$54 000
\$73 000 to \$83 000	\$2200
\$83 000 to \$94 000	\$2200 less 20 cents for each \$1 that annual income exceeds \$83 000
Over \$94 000	Nil

Families with 2 or more children meeting the criteria	
Combined annual income	**Allowance for the year**
Up to \$54 000	\$11 100
\$54 000 to \$73 000	\$11 100 less 20 cents for each \$1 that annual income exceeds \$54 000
\$73 000 to \$97 000	\$7300
\$97 000 to \$110 000	\$7300 less 30 cents for each \$1 that annual income exceeds \$97 000
\$110 000 to \$130 000	\$4400
\$130 000 to \$140 000	\$4400 less 44 cents for each \$1 that annual income exceeds \$130 000
Over \$140 000	Nil

(a) Determine the allowance paid to a family with one child and a combined annual income of \$76 000. (1 mark)

Solution
- \$2200
Specific behaviours
✓ correct allowance

(b) Determine the allowance paid to a family with two children and a combined annual income of \$105 000. (2 marks)

Solution
-\$7300 – (105 000 – 97 000) x 0.3 = \$4900
Specific behaviours
✓ uses correct allowance, (7300 and 97 000 shown in calc.) ✓ correct answer given.

(c) Determine the allowance paid to a family with no children and a combined annual income of \$83 000. (1 mark)

Solution
- Nil
Specific behaviours
✓ No allowance given

(d) Determine the combined income a family with one child would earn if the allowance paid to them was \$4600. (2 marks)

Solution
4600 = 5100 – (x – 54 000) x 0.1 Combined income of \$59 000
Specific behaviours
✓ Shows calculation using 4600 and 54 000 ✓Correct combined income shown.

(e) Determine the possible minimum and maximum combined income a family with 6 children could earn if the allowance paid to them was \$7300. (2 marks)

Solution
Minimum \$73 000 Maximum \$97 000
Specific behaviours
✓ Shows Min. ✓ Shows Maximum.

Question 21 **(8 marks)**

The washer shown below is made of plastic and has an internal radius of 7 mm, an external radius of 22 mm and is 3 mm thick.

{Washer Drawing}

(a) The plastic used to make the washers costs 95 cents per litre. Given that one litre is 1 000 cubic centimetres, determine the cost of making 8 000 of the washers. (4 marks)

Solution
Outer volume: $\pi \times 2.2^2 \times 0.3 = 4.562$ Inner volume: $\pi \times 0.7^2 \times 0.3 = 0.462$ Total volume: $(4.562 - 0.462) \times 8000 = 4.1 \times 8000 = 32800$ Cost: $32800 \div 1000 \times 0.95 = \$31.16$
Specific behaviours
✓ outer volume ✓ inner volume ✓ volume of all washers ✓ correct cost

(b) Calculate the total surface area of one washer, giving your answer in mm². (4 marks)

Solution
Outer cylinder: $A_1 = 2\pi(22)^2 + 2\pi(22)(3) = 3456$ Inner circles: $A_2 = 2\pi(7)^2 = 308$ Inner walls: $A_3 = 2\pi(7)(3) = 132$ TSA: $A = 3456 - 308 + 132 = 3\,280$ mm²
Specific behaviours
✓ outer cylinder ✓ inner circles ✓ inner walls ✓ correct TSA

Alternative Solution
Top Face: $= 2\pi(22)^2 - 2\pi(7)^2 = 1366.59$ Top & Bottom Face $A_1 = (2)(1366.59) = 2733.19$ Inner wall: $A_2 = 2\pi(7)(3) = 131.95$ Outer walls: $A_3 = 2\pi(22)(3) = 414.69$ TSA: $A = 2733.19 + 131.95 + 414.69 = 3\,280$ mm²
Specific behaviours
✓ One Area (Top/Bottom) ✓ inner wall area ✓ Outer walls area ✓ correct TSA

Supplementary page

Question number: _________

Supplementary page

Question number: _________